

PRAKTIKUM SISTEM OPERASI

MODUL 6
INSTALASI XINU



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
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1. **[10 Poin]** Selain hardware (memory), batasan maksimal proses dapat ditentukan dengan secara software. Pada Linux maksimal proses adalah 4194303 proses (64 bit) dan 32767 proses (32 bit) dapat dilihat melalui perintah `$cat /proc/sys/kernel/pid_max`

Carilah pada source code Xinu yang memberi batasan mengenai banyaknya proses yang bisa dibuat! Berapa maksimal proses dalam Xinu? Ubah menjadi maksimal 150 proses!

- a. Membuka file `conf.h`, lalu merubah maksimal proses dari 100 menjadi 150

```
Activities Terminal Mon 15 Jun, 19:14
xinu@xinu-develop-end: ~/xinu/include
File Edit View Search Terminal Help
GNU nano 2.2.6 File: conf.h Modified

#define Nlfs 1
#define Nlfl 6
#define Nnam 1
#define Npipem 1
#define Npip 10

#define NDEVS 35

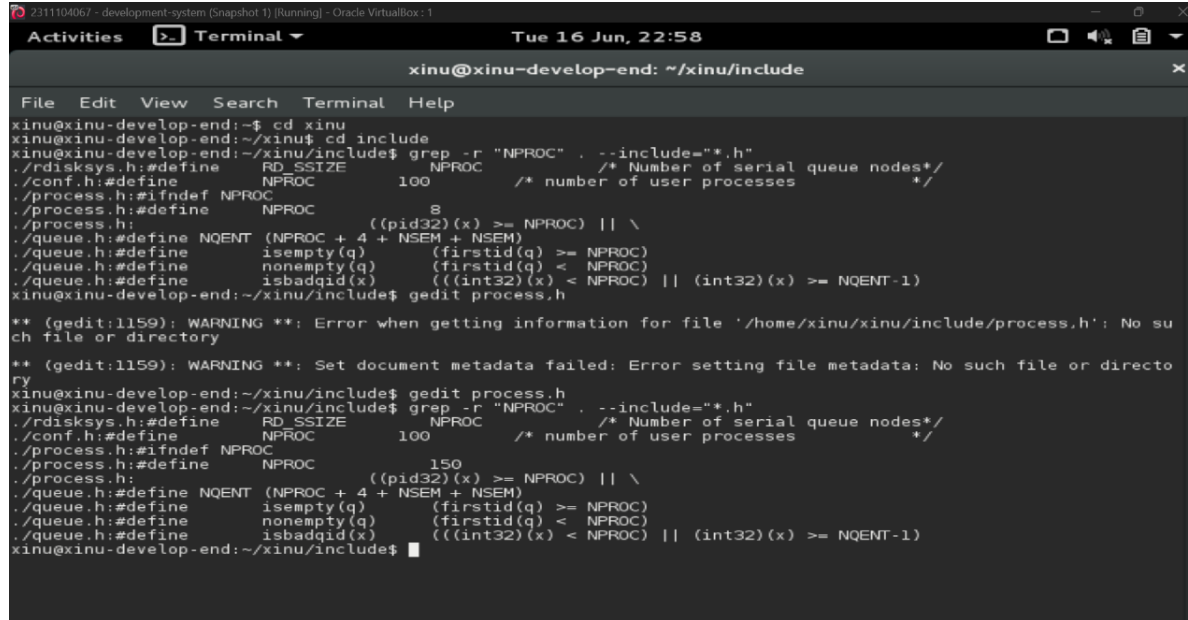
/* Configuration and Size Constants */

#define NPROC 100 /* number of user processes */
#define NSEM 100 /* number of semaphores */
#define IRQBASE 32 /* base ivec for IRQ0 */
#define IRQ_TIMER IRQ_HW5 /* timer IRQ is wired to hardware 5 */
#define IRQ_ATH_MISC IRQ_HW4 /* Misc. IRQ is wired to hardware 4 */
#define CLKFREQ 200000000 /* 200 MHz clock */

#ifndef ETHER0
#define ETHER0 0
#endif

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

- b. Verivikasi Perubahan nilai NPORC



The screenshot shows a terminal window titled "xinu@xinu-develop-end: ~/xinu/include". The user is editing the file `process.h` using `gedit`. The code defines `NPROC` as 100 and `NQENT` as `(NPROC + 4 + NSEM + NSEM)`. The user has changed the value of `NPROC` from 100 to 150. The terminal output shows the following commands and their results:

```
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ cd include
xinu@xinu-develop-end:~/xinu/include$ grep -r "NPROC" . --include="*.h"
./rdisksys.h:#define RD_SSIZE NPROC /* Number of serial queue nodes*/
./conf.h:#define NPROC 100 /* number of user processes */
./process.h:#ifndef NPROC
./process.h:#define NPROC 8
./process.h:((pid32)(x) >= NPROC) || \
./queue.h:#define NQENT (NPROC + 4 + NSEM + NSEM)
./queue.h:#define isempty(q) (firstid(q) >= NPROC)
./queue.h:#define nonempty(q) (firstid(q) < NPROC)
./queue.h:#define isbadqid(x) (((int32)(x) < NPROC) || ((int32)(x) >= NQENT-1))
xinu@xinu-develop-end:~/xinu/include$ gedit process.h

** (gedit:1159): WARNING **: Error when getting information for file '/home/xinu/xinu/include/process.h': No such file or directory
** (gedit:1159): WARNING **: Set document metadata failed: Error setting file metadata: No such file or directory
xinu@xinu-develop-end:~/xinu/include$ gedit process.h
xinu@xinu-develop-end:~/xinu/include$ grep -r "NPROC" . --include="*.h"
./rdisksys.h:#define RD_SSIZE NPROC /* Number of serial queue nodes*/
./conf.h:#define NPROC 100 /* number of user processes */
./process.h:#ifndef NPROC
./process.h:#define NPROC 150
./process.h:((pid32)(x) >= NPROC) || \
./queue.h:#define NQENT (NPROC + 4 + NSEM + NSEM)
./queue.h:#define isempty(q) (firstid(q) >= NPROC)
./queue.h:#define nonempty(q) (firstid(q) < NPROC)
./queue.h:#define isbadqid(x) (((int32)(x) < NPROC) || ((int32)(x) >= NQENT-1))
xinu@xinu-develop-end:~/xinu/include$
```

2. [20 Poin] Jalankan kode main_ex1!

- a. Mengubah isi code pada file main.c

```
/* main.c - main */

#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void)
{
    sndA();
    sndB();
    return OK;
}

void sndA(void)
{
    while(1){
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1){
        printf("B\n");
        sleepms(1000);
    }
}
```

- b. Hasil compile

```
Obtained IP address 10.0.3.15    (0xa00030f)
A
A
A
A
A
A
A
A
A
A
A
A
A
A
A
A
A
A
```

3. **[20 Poin]** Jalankan kode main_ex2!

```

/* main.c - main */

#include <xinu.h>
void sndA(void);
void sndB(void);

process main(void)
{
    resume(create(sndA, 1024, 20, "process A", 0));
    resume(create(sndB, 1024, 20, "process B", 0));
    return OK;
}

void sndA(void)
{
    while(1){
        printf("A\n");
        sleepms(1000);
    }
}

void sndB(void)
{
    while(1){
        printf("B\n");
        sleepms(1000);
    }
}

```

Hasil compile:

```
Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B
A
B
A
B
```

4. **[20 Poin]** Jalankan kode main_ex3!

Mengganti code di file main:

```

/* main.c - main*/

#include <xinu.h>
void sndChar(char);

process main(void)
{
    resume(create(sndChar, 1024, 20, "Send A", 1, 'A'));
    resume(create(sndChar, 1024, 20, "Send B", 1, 'B'));
    return OK;
}

void sndChar(char c)
{
    while(1){
        printf("%c \n", c);
        sleepms(1000);
    }
}

```

Hasil compile:

```
Obtained IP address 10.0.3.15 (0x0a00030f)
A
B
A
B
A
B
A
B
B
A
B
B
A
A
```

5. **[30 Poin]** Buatlah 2 proses produser dan konsumen. Produser memproduksi angka integer dari 1-1000. Konsumen mengkonsumsi integer yang diproduksi oleh produser dan menampilkannya! (Gunakan variabel global bertipe int32 bernama n yang digunakan secara bersama oleh kedua proses)

```

/* main.c - main*/

#include <xinu.h>

int32 n = 0;

void producer(void) {
    int32 i;
    for (i = 1; i <= 1000; i++) {
        n++;
    }
}

void konsumer(void) {
    int32 i;
    for (i = 1; i <= 1000; i++) {
        printf("Nilai dari n adalah %d\n", n);
    }
}

process main(void) {
    resume(create(producer, 1024, 20, "producer", 0));
    resume(create(konsumer, 1024, 20, "konsumer", 0));
    return OK;
}

```

Hasil compile:

[illegible]

